

Macroeconomics III

Prof. Guido Cozzi

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Chapter 4

WEALTH
ACCUMULATION AND
CAPITAL MOBILITY:
THE SOLOW MODEL
FOR A SMALL OPEN
ECONOMY

INTRODUCTION

- In a closed economy,

$$S_t - I_t = 0.$$

The only source of domestic investment is domestic savings. Provided international capital flows are small, this is a good approximation.

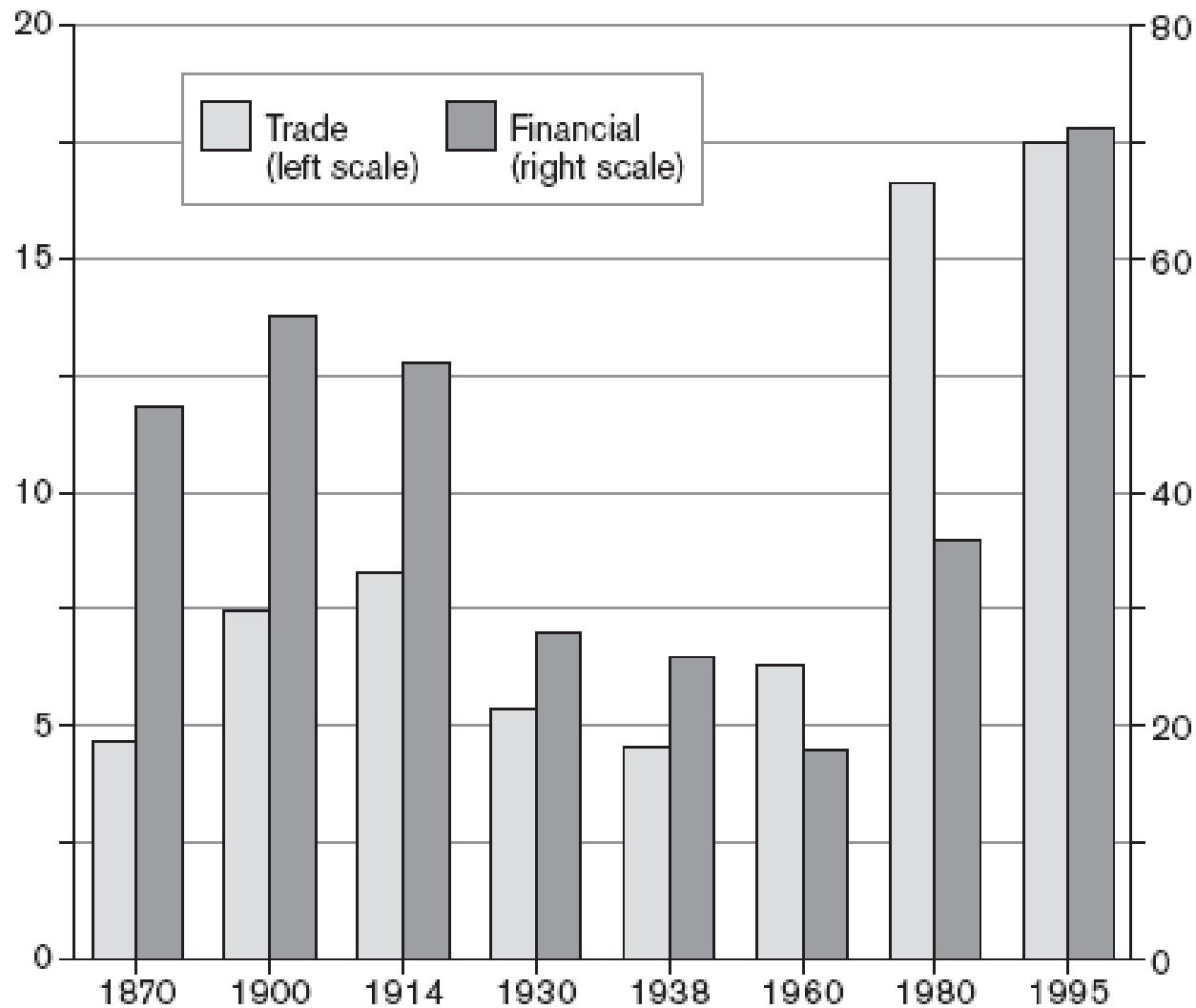
- In an open economy,

$$S_t - I_t = \text{capital export} = \text{current account balance}.$$

If domestic investment is larger than domestic saving, there is a positive net import of capital and a deficit on the current account.

- Since 1960, capital mobility and capital flows have increased dramatically. This is one aspect of globalization:

Foreign trade and foreign assets relative to world GDP



- Therefore, we will study a Solow model that takes capital movements into account:
 - Focus on wealth (versus capital) accumulation and capital movements
 - Not focus on commodity trade (this is not trade theory)
 - Still not focus on technological progress
- Main question: is free mobility of capital (and hence liberalization of capital movements) good? (Not: "is trade good?")

CAPITAL MOVEMENTS, SAVINGS AND INVESTMENT

- In the closed economy the national intertemporal budget constraint was: $K_{t+1} - K_t + \delta K_t = S_t$. In the open economy it is:

$$F_{t+1} - F_t + \overbrace{K_{t+1} - K_t + \delta K_t}^{=I_t} = S_t \Leftrightarrow$$

$$F_{t+1} - F_t = S_t - I_t \quad (*)$$

where F_t is net foreign assets.

- The goods markets equilibrium/national accounting identity is

$$Y_t + M_t = C_t + I_t + X_t \Leftrightarrow$$

$$\overbrace{Y_t + \bar{r}F_t - C_t}^{S_t} - I_t = \overbrace{X_t - M_t + \bar{r}F_t}^{CA_t} \quad (**)$$

where $\bar{r}$ = the international real interest rate. Combining (*) and (**) gives:

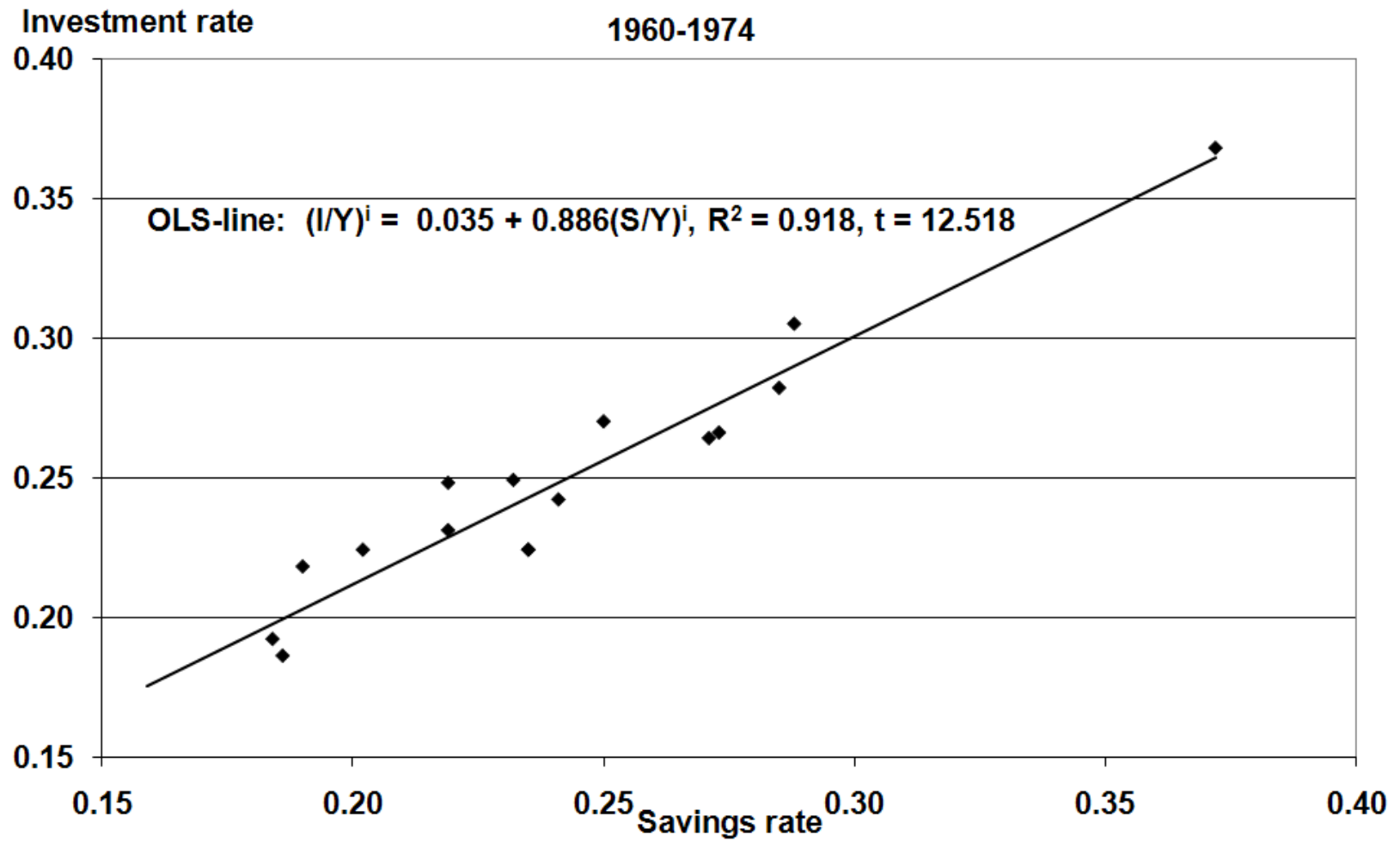
$$S_t - I_t = F_{t+1} - F_t = CA_t$$

- Without capital movements (e.g., in a closed economy):
 $F_{t+1} = F_t = 0 \Leftrightarrow$ and $S_t / Y_t = I_t / Y_t$.
- Idea: measure the degree of capital mobility by checking how far away from "perfect" the correlation between S_t / Y_t and I_t / Y_t is.
- To check this, perform an OLS estimation of

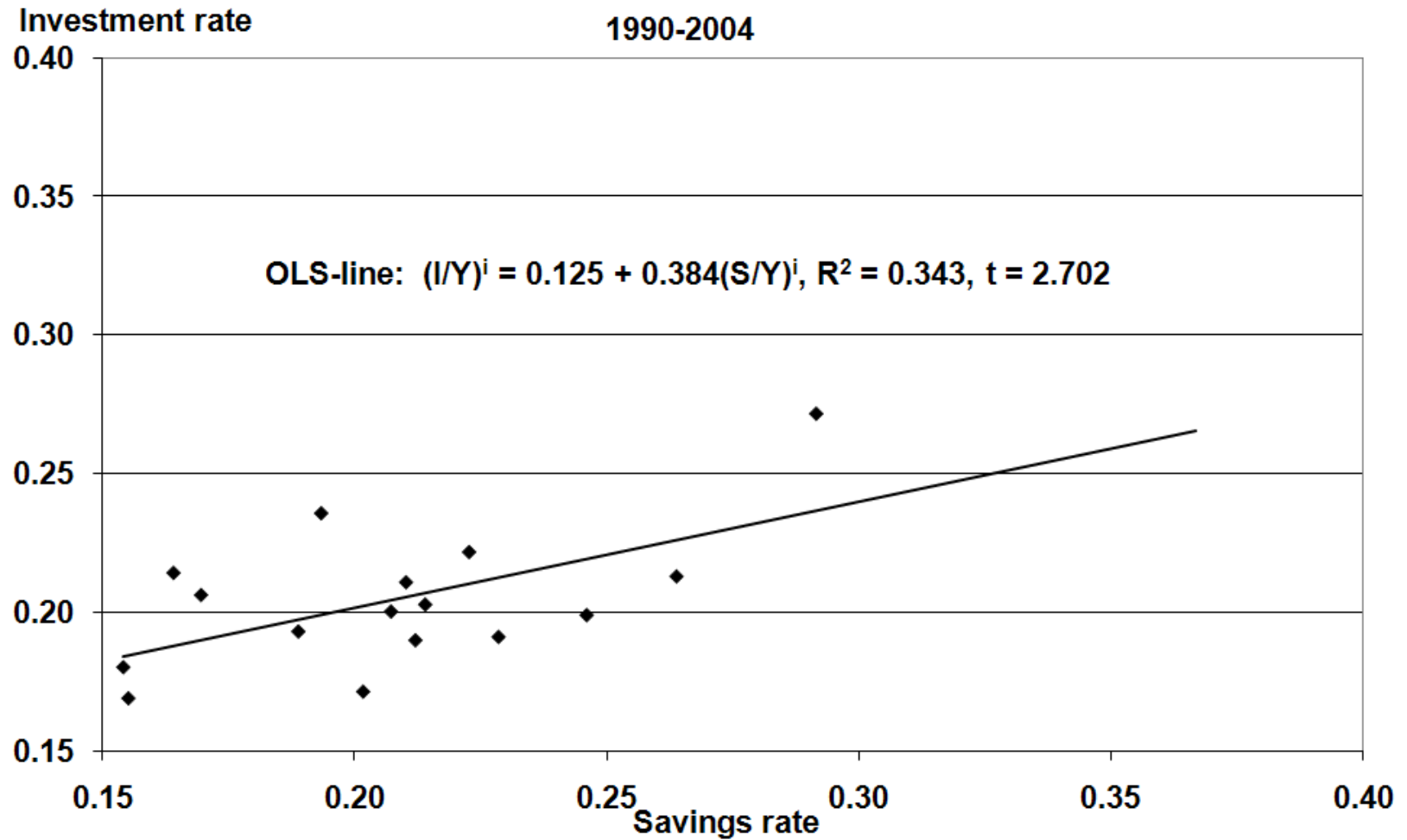
$$\left(\frac{I_t}{Y_t} \right)^i = \alpha_0 + \alpha_1 \left(\frac{S_t}{Y_t} \right)^i$$

where i refers to country i . For 16 OECD countries for each of the two subperiods 1960-1974 and 1990-2004, this gives a clear tendency to a less tight correlation in the second subperiod:

National saving and national investment in 16 OECD countries, 1960-1974



National saving and national investment in 16 OECD countries, 1990-2004



Crucial Questions

The crucial **new questions** are:

- Does an open capital account benefit the economy in the long run?
- Do new risks come up with free capital mobility?

SOLOW MODEL FOR A SMALL OPEN ECONOMY

... with perfect mobility of capital and goods

- **Open:** Goods and capital can move freely over country borders. No mobility of labour!
- **Small:** The economy considered has negligible influence on the World economy. Specifically, domestic investment and savings have no effect on the world interest rate.
- **Perfect mobility of capital:** No barriers for capital movements $\Rightarrow$ real return on domestic capital equals real return on investment abroad. That is, the domestic real interest rate is determined by the world real interest rate.
- **Perfect mobility of goods:** No barriers for exports and imports, but since we assume that all countries produce the same single good, movements of goods only reflect capital movements; there is no explicit theory of trade patterns or gains from trade.

THE MODEL

- Simplifying assumption: $\delta = 0$.
- There is a distinction between domestic capital, K_t , and domestic (national) wealth, V_t . The difference is domestic net foreign assets, F_t :

$$V_t = K_t + F_t.$$

Note: If all or part of K_t is owned by foreigners it will enter negatively in F_t .

- Intertemporal budget constraint (remember $\delta = 0$):

$$V_{t+1} - V_t = S_t$$

- Capital invested in the international market yields a constant real rate of return, $\bar{r} > 0$. If Y_t is domestic production and income (GDP), national income (GNP) is

$$Y_t^n = Y_t + \bar{r}F_t.$$

- As in the case of the basic Solow model, aggregate domestic product is given by:

$$Y_t = BK_t^\alpha L_t^{1-\alpha}$$

National savings are assumed to be a constant fraction, s , of national income:

$$S_t = sY_t, \quad 0 < s < 1$$

- As usual the labour force evolves as

$$L_{t+1} = (1+n)L_t, \quad n > -1$$

- Profit maximization implies that the marginal products of capital and labour equal the real factor prices, r_t and w_t , respectively, where r_t is the domestic real interest rate (recall $\delta = 0$). Given that $r_t = \bar{r}$ we have that

$$\bar{r} = \alpha B \left(\frac{K_t}{L_t} \right)^{\alpha-1} \quad \text{and} \quad w_t = (1-\alpha) B \left(\frac{K_t}{L_t} \right)^{\alpha}$$

THE COMPLETE SOLOW MODEL OF THE OPEN ECONOMY

$$V_t = K_t + F_t$$

$$V_{t+1} - V_t = S_t, \quad V_0 \text{ given}$$

$$Y_t^n = Y_t + \bar{r}F_t$$

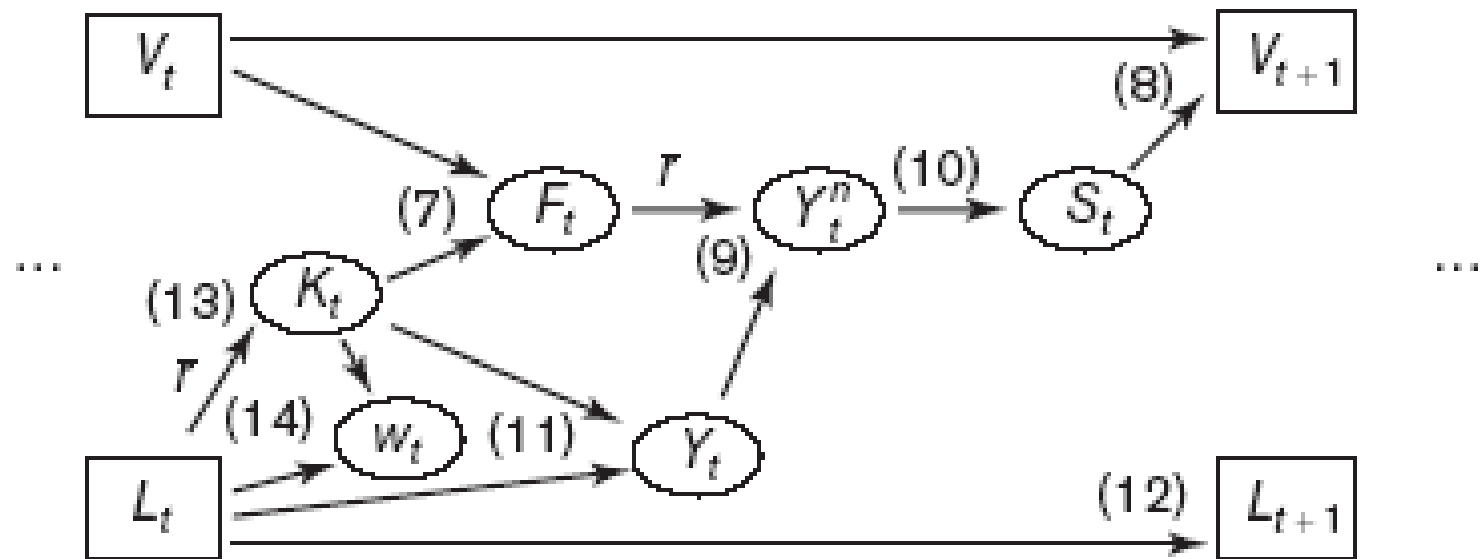
$$Y_t = BK_t^\alpha L_t^{1-\alpha}$$

$$L_{t+1} = (1+n)L_t, \quad L_0 \text{ given}$$

$$S_t = sY_t^n$$

$$\bar{r} = \alpha B \left(\frac{K_t}{L_t} \right)^{\alpha-1} \quad \text{and} \quad w_t = (1-\alpha) B \left(\frac{K_t}{L_t} \right)^\alpha$$

The dynamic links of the Solow model



ANALYSING THE MODEL

1. Per capita (worker) variables:

$$y_t \equiv \frac{Y_t}{L_t}, \quad k_t \equiv \frac{K_t}{L_t}, \quad y_t^n \equiv \frac{Y_t^n}{L_t}, \quad v_t \equiv \frac{V_t}{L_t}, \quad f_t \equiv \frac{F_t}{L_t}.$$

$$v_t = k_t + f_t, \quad y_t^n = y_t + \bar{r}f_t, \quad y_t = Bk_t^\alpha.$$

2. Income per capita, y_t^n :

$$\bar{r} = \alpha Bk_t^{\alpha-1} \Leftrightarrow \bar{r}k_t = \alpha Bk_t^\alpha = \alpha y_t,$$

$$w_t = (1 - \alpha) Bk_t^\alpha = (1 - \alpha) y_t.$$

Hence $\bar{r}k_t + w_t = y_t$, $\bar{r}k_t / y_t = \alpha$ and $w_t / y_t = 1 - \alpha$, etc.

$$y_t^n = y_t + \bar{r}f_t = y_t + \bar{r}(v_t - k_t) = y_t - \bar{r}k_t + \bar{r}v_t = w_t + \bar{r}v_t$$

National income per capita equals wage plus return (real interest rate) to national wealth per capita.

THE LAW OF MOTION

3. Immediate adjustment: Through import/export of capital, the domestic capital intensity, k_t , adjusts in each period to accommodate $\bar{r} = \alpha B k_t^{\alpha-1} \Leftrightarrow$

$$k_t = k^* = B^{\frac{1}{1-\alpha}} \left(\frac{\alpha}{\bar{r}} \right)^{\frac{1}{1-\alpha}}.$$

Correspondingly, y_t and w_t jump immediately to:

$$y^* = B^{\frac{1}{1-\alpha}} \left(\frac{\alpha}{\bar{r}} \right)^{\frac{\alpha}{1-\alpha}}, \quad w^* = (1-\alpha) B^{\frac{1}{1-\alpha}} \left(\frac{\alpha}{\bar{r}} \right)^{\frac{\alpha}{1-\alpha}}$$

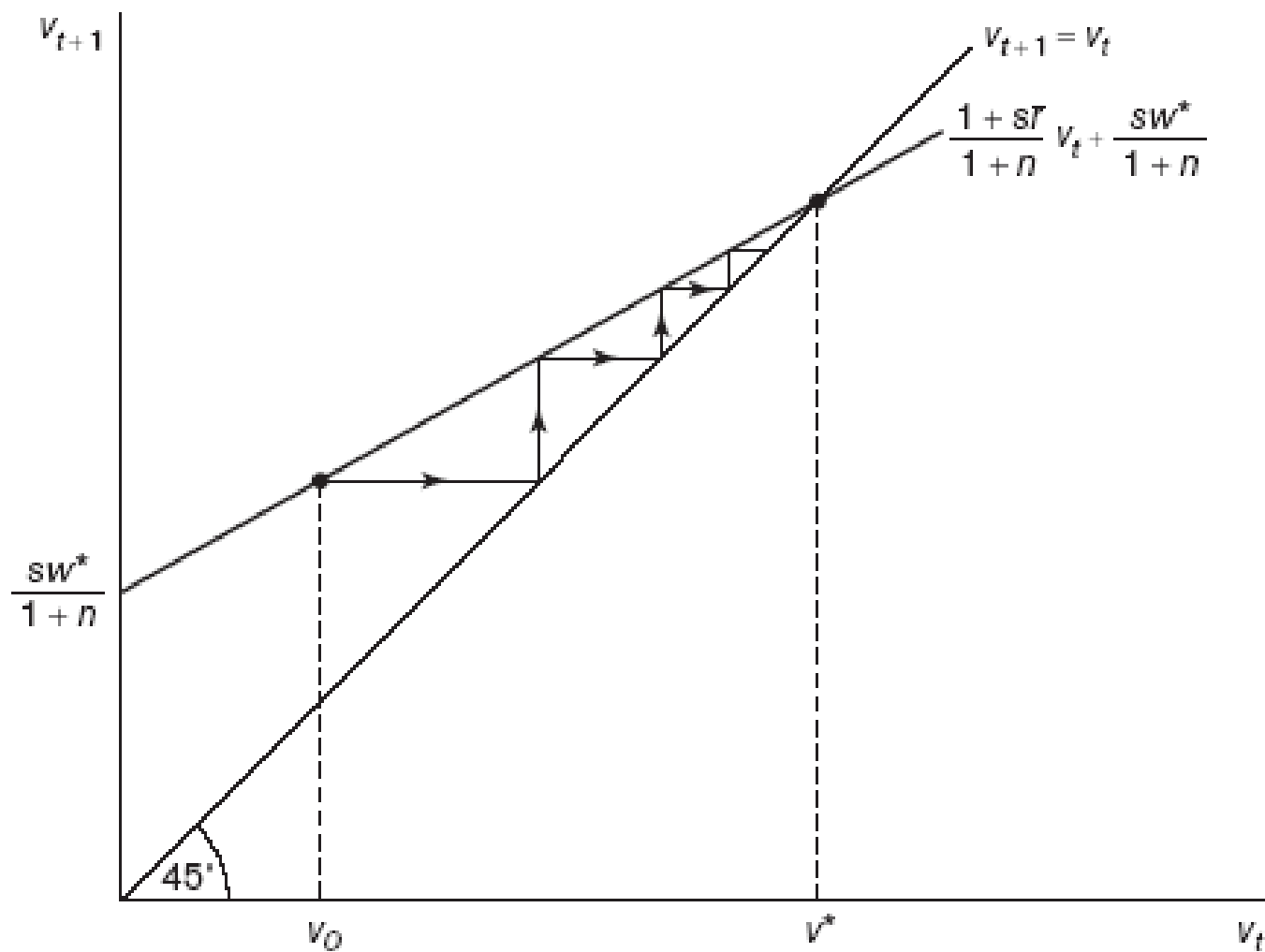
4. Gradual adjustment: Wealth accumulates gradually:

$$\begin{aligned} V_{t+1} &= S_t + V \Rightarrow \\ v_{t+1} &= \frac{1}{1+n} \left(v_t + \frac{sY_t^n}{L_t} \right) = \frac{1}{1+n} (v_t + sy_t^n) \\ &= \frac{1}{1+n} \left(v_t + s(w^* + \bar{r}v_t) \right) \Leftrightarrow \\ v_{t+1} &= \frac{1+s\bar{r}}{1+n} v_t + \frac{sw^*}{1+n} \end{aligned}$$

This is the **transition equation**. It determines the sequence (v_t) for a given v_0 and thus (y_t^n) and (c_t) are determined through $y_t^n = w^* + \bar{r}v_t$ and $c_t = (1-s)y_t^n$, etc.

The transition diagram

$$v_{t+1} = \frac{1 + s\bar{r}}{1 + n} v_t + \frac{sw^*}{1 + n}$$



Stability condition:

$$s\bar{r} < n$$

Is this a reasonable assumption? Yes, from the following argument:

In a closed economy, $r_c^* = \alpha n / s$. [Prove it!]

Since the world economy is closed, $\bar{r} = \alpha \bar{n} / \bar{s}$.

Insert this into the stability condition to find that

$$s\bar{r} < n \Leftrightarrow \alpha \frac{s}{n} < \frac{\bar{s}}{\bar{n}}$$

$$\alpha \approx 1/3.$$

which is highly plausible.

KEY VARIABLES IN STEADY STATE

$$v_{t+1} = \frac{1 + s\bar{r}}{1 + n} v_t + \frac{sw^*}{1 + n}$$

1. Wealth per capita. In the transition equation, set

$v_{t+1} = v_t = v^*$ to get:

$$v^* = \frac{s}{n - s\bar{r}} w^* = \frac{s/n}{1 - s\bar{r}/n} w^*, \text{ where } \frac{s\bar{r}}{n} < 1$$

2. Net foreign asset position: Remember that $f^* = v^* - k^*$. We just found v^* and, from our formulas for k^* and w^* follows $k^* = \frac{1}{1 - \alpha} \frac{\alpha}{\bar{r}} w^*$. Combining these three elements gives:

$$f^* = \frac{1}{1 - \alpha} \frac{s}{n} \frac{1}{\bar{r}} \frac{\bar{r} - \alpha n / s}{1 - s\bar{r}/n} w^*$$

It follows that:

$$f^* > 0 \Leftrightarrow \bar{r} > \frac{\alpha n}{s} \Leftrightarrow \bar{r} > r_c^*$$

A country becomes a net creditor if the domestic real interest rate that it would create under autarky is smaller than the international real interest rate.

Or, using again that $\bar{r} = \alpha \bar{n} / \bar{s}$, if

$$s / n > \bar{s} / \bar{n}$$

that is, if the domestic economy has a relatively strong propensity to save.

3. Remember: Capital per capita, GDP per capita and the real wage rate jump to k^* , y^* and w^* , respectively.

4. **National income per capita.** $y^{n*} = w^* + \bar{r}v^* \Rightarrow$

$$y^{n*} = w^* + \bar{r} \frac{S}{n - s\bar{r}} w^* = w^* \frac{n}{n - s\bar{r}} = w^* \frac{1}{1 - s\bar{r} / n}.$$

Inserting our expression for w^* and rearranging gives:

$$y^{n*} = (1 - \alpha) B^{\frac{1}{1-\alpha}} \left(\frac{\alpha}{\bar{r}} \right)^{\frac{\alpha}{1-\alpha}} \frac{1}{1 - s\bar{r} / n}$$

and c^* follows from $c^* = (1 - s) y^{n*}$.

Implications for structural policies: A high value of B and of s / n tends to create high values of consumption and income per capita. This is more or less as in the closed economy. The quantitative effects are even of similar size.

The crucial **new questions** are:

- Does an open capital account benefit the economy in the long run?
- Do new risks come up with free capital mobility?

ADVANTAGES OF FREE CAPITAL MOBILITY?

- **Method:** Compare the long run levels of national income per capita and other relevant variables under autarky with the same variables under free capital mobility.
- **Result:** Steady state national income per capita is always higher under free capital mobility. But the real wage is lower if the country has a relatively strong propensity to save.

Intuition for this result

If the economy were closed, then $r_c^* = \alpha n / s$, and similarly the world real interest rate is $\bar{r} = \alpha \bar{n} / \bar{s}$. Two possibilities:

1. $r_c^* < \bar{r}$ or 2. $r_c^* > \bar{r}$

1. The economy has a relatively strong propensity to save ($s / n > \bar{s} / \bar{n}$) $\Rightarrow$ capital accumulation is relatively strong and the real interest rate is relatively low under autarky.

Opening the capital account: the owners of capital take advantage of a higher international interest rate $\Rightarrow$ capital flows out of the domestic economy $\Rightarrow$ capital intensity decreases $\Rightarrow$ higher MP of capital, lower MP of labour $\Rightarrow$ domestic interest rate increases to r_c^* , domestic real wages fall.

National income increases because national wealth is invested at a higher interest rate. The owners of capital profit from the opening and real wages fall. The country becomes a net creditor.

2. The economy has a relatively low propensity to save ($s / n < \bar{s} / \bar{n}$) $\Rightarrow$ capital accumulation is relatively weak and the real interest rate is relatively high under autarky.

An opening creates an arbitrage possibility: borrow on the international capital market and invest in domestic capital, capital flows in $\Rightarrow$ capital intensity increases $\Rightarrow$ MP of capital decreases, MP of labour increases $\Rightarrow$ domestic interest rate falls down to r_c^* and domestic real wage increases.

National income increases because of the advantageous arbitrage in which the economy takes part. The capital owners get a lower rate of return (but gain from the arbitrage), the real wage increases and the economy becomes a net debtor.

Regardless of the initial situation: Opening the capital account increases long term national income per capita!

Formal proof of $y^{n*} > y_c^*$

- Use the above expression for national income in the open economy and take the analogous expression for the closed economy from chapter 3 (now with $\delta = 0$):

$$y^{n*} = (1 - \alpha) B^{\frac{1}{1-\alpha}} \left(\frac{\alpha}{\bar{r}} \right)^{\frac{\alpha}{1-\alpha}} \frac{1}{1 - s\bar{r} / n} \quad \text{and} \quad y_c^* = B^{\frac{1}{1-\alpha}} \left(\frac{s}{n} \right)^{\frac{\alpha}{1-\alpha}}$$

- Define:

$$x = \frac{y^{n*}}{y_c^*} = (1 - \alpha) \left(\frac{1}{\bar{r}} \frac{\alpha n}{s} \right)^{\frac{\alpha}{1-\alpha}} \frac{1}{1 - s\bar{r} / n}$$

- Use that $r_c^* = \alpha n / s \Rightarrow s / n = \alpha / r_c^*$ to get

$$\begin{aligned} x &= (1 - \alpha) \left(\frac{r_c^*}{\bar{r}} \right)^{\frac{\alpha}{1-\alpha}} \frac{1}{1 - \alpha \bar{r} / r_c^*} \\ &= (1 - \alpha) \left(\frac{1}{\tilde{r}} \right)^{\frac{\alpha}{1-\alpha}} \frac{1}{1 - \alpha \tilde{r}}, \quad \tilde{r} \equiv \frac{\bar{r}}{r_c^*}. \end{aligned}$$

- Three properties of x as a function of $\tilde{r}$:

1. If $\tilde{r} = 1$ ($\bar{r} = r_c^*$) then $x = 1 \Leftrightarrow y^{n*} = y_c^*$.

2. The derivative:

$$\frac{d \ln x}{d \tilde{r}} = -\frac{\alpha}{1 - \alpha} \frac{1}{\tilde{r}} + \frac{\alpha}{1 - \alpha \tilde{r}}$$

is zero for $\tilde{r} = 1$.

3. The same derivative is increasing in $\tilde{r}$, as long as $\alpha\tilde{r} < 1$ (the stability condition).

- Consequently, $x = y^{n*} / y_c^*$ has a global minimum for $\tilde{r} = 1$ where $y^{n*} = y_c^*$. For all other values of $\tilde{r}$ we have that $y^{n*} > y_c^*$. End of proof.

Formal proof of real wage statement

- Look at the expression for the real wage in the open and in the closed economy with ($\delta = 0$):

$$w^* = (1 - \alpha) B^{\frac{1}{1-\alpha}} \left(\frac{\alpha}{\bar{r}} \right)^{\frac{\alpha}{1-\alpha}} \quad \text{and} \quad w_c^* = (1 - \alpha) B^{\frac{1}{1-\alpha}} \left(\frac{s}{n} \right)^{\frac{\alpha}{1-\alpha}} \Rightarrow$$

$$\frac{w}{w_c} = \left(\frac{\alpha n / s}{\bar{r}} \right)^{\frac{\alpha}{1-\alpha}} = \left(\frac{r_c}{\bar{r}} \right)^{\frac{\alpha}{1-\alpha}}$$

When $r_c^* > \bar{r}$ (the economy has a relatively weak propensity to save, $s / n < \bar{s} / \bar{n}$), we have $w^* > w_c^*$ and vice versa.

NEW RISKS?

- A closed economy does not expose itself to interest rate shocks. What happens if the economy is in steady state initially and $\bar{r}$ increases permanently? (No formal analysis)
- Capital flows out of the country $\Rightarrow$ lower capital intensity $\Rightarrow$ lower real wage, lower GDP per capita and higher return on domestic capital. Furthermore, the return on national wealth increases (remember that $y^n = w + \bar{r}v$).
- If the country is a net debtor initially then y^n decreases, if it is a net creditor y^n increases, etc.:

	creditor	debtor
$\bar{r}$ rises	y^n rises w falls	y^n falls w falls
$\bar{r}$ falls	y^n falls w rises	y^n rises w rises

- If a country wants to avoid these “ups and downs”, it can choose to close its capital account.

CONCLUSION

- Implications for domestic structural policy intended to increase income and consumption per capita in the long run are qualitatively the same as in the Solow model of a closed economy, i.e., increase savings rate, etc. The effects of such policies are even quantitatively of similar magnitude.
- Should capital movements be liberalized?
It has an unambiguously positive effect on national income per capita in the long run but ambiguous effects on factor incomes.

Negative implications:

- particular factor income (e.g., real wage) is sometimes affected negatively
- the economy becomes exposed to interest rate shocks

CONCLUSION

- The international trend is liberalization. However, second thoughts have begun to emerge because in several cases capital has flown out of, not into, poor countries upon liberalization. The risk to the country can be very important!
- The positive welfare effects of full opening may also depend on the degree of rationality of markets, which cannot be taken for granted.¹

1. For research on this issue at this Chair, see Cozzi and Davenport "[Extrapolative Expectations and Capital Flows during Convergence](#)" (2017).